Math 242, S20 **Name:** Answer

Quiz 9

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find the Laplace transforms of the first two functions and the inverse Laplace transforms of the last two functions:

(a) $f(t) = -2e^{3t}$

$$\mathcal{L}\{-2e^{3t}\} = -2\mathcal{L}\{e^{3t}\}$$
$$= -\frac{2}{s-3}, \quad s > 3$$

(b) $f(t) = \sin 3t + 2\cos 3t$

$$\mathcal{L}\{\sin 3t + 2\cos 3t\} = \mathcal{L}\{\sin 3t\} + 2\mathcal{L}\{\cos 3t\}$$
$$= \frac{3}{s^2+9} + 2\frac{s}{s^2+9}$$
$$= \frac{2s+3}{s^2+9}, \quad s > 0$$

(c) $F(s) = \dfrac{3}{s^4}$

$$\mathcal{L}^{-1}\left\{\frac{3}{s^4}\right\} = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{3!}{s^4}\right\}$$
$$= \frac{1}{2}t^3$$

(d) $F(s) = \dfrac{4}{s^2-4}$

$$\mathcal{L}^{-1}\left\{\frac{4}{s^2-4}\right\} = \mathcal{L}^{-1}\left\{\frac{2\cdot 2}{s^2-2^2}\right\}$$
$$= 2\mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$$
$$= 2\sinh(2t)$$

Problem 2. (5 points) Use Laplace transform to solve the initial value problem:

$$x'' - 2x' - 3x = 0; \quad x(0) = 3, \; x'(0) = 7.$$

Let $X(s) = \mathcal{L}\{x(t)\}$, we have

$$\mathcal{L}\{x'(t)\} = s\mathcal{L}\{x(t)\} - x(0)$$
$$= sX(s) - 3$$
$$\mathcal{L}\{x''(t)\} = s^2\mathcal{L}\{x(t)\} - sx(0) - x'(0)$$
$$= s^2X(s) - 3s - 7$$

thus

$$\mathcal{L}\{x'' - 2x' - 3x\} = \mathcal{L}\{0\}$$
$$\mathcal{L}\{x''\} - 2\mathcal{L}\{x'\} - 3\mathcal{L}\{x\} = 0$$
$$[s^2X(s) - 3s - 7] - 2[sX(s) - 3] - 3X(s) = 0$$

$$(s^2 - 2s - 3)X(s) = 3s + 1$$
$$\Rightarrow X(s) = \frac{3s+1}{s^2-2s-3}$$

$$\frac{3s+1}{s^2-2s-3} = \frac{3s+1}{(s-3)(s+1)} = \frac{A}{s-3} + \frac{B}{s+1}$$
$$= \frac{A(s+1) + B(s-3)}{(s-3)(s+1)} = \frac{(A+B)s + (A-3B)}{(s-3)(s+1)}$$

$$\Rightarrow \begin{cases} A + B = 3 \\ A - 3B = 1 \end{cases} \Rightarrow \begin{array}{l} A = \frac{5}{2} \\ B = \frac{1}{2} \end{array}$$

thus

$$x(t) = \mathcal{L}^{-1}\left\{\frac{3s+1}{s^2-2s-3}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{5}{2}\frac{1}{s-3}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{2}\frac{1}{s+1}\right\}$$
$$= \frac{5}{2}e^{3t} + \frac{1}{2}e^{-t}$$